Flutter Cheat Sheet

Flutter 3.47.9 · Dart 3.13.7 · go_router 18.1.4 · flutter_riverpod 3.5.0 · dio 5.11.1 · AI-generated summary — verify against the official docs · see the Flutter pages for details

CLI — PROJECT

Set up and create

```
flutter doctor       # check the toolchain
flutter create trailhead
flutter devices      # devices, emulators
```

Dependencies

```
flutter pub add dio
flutter pub add dev:mocktail   # dev dep
flutter pub get      # from pubspec.yaml
```

Run, analyse, generate

```
flutter run -d <device-id>
# r = hot reload, R = hot restart
flutter analyze
dart format .
dart run build_runner build
# json_serializable, drift generators
```

DART — ESSENTIALS

Variables and null safety

```
final name = 'Ridge'; var count = 0;
const pi = 3.14;
String? note; note?.length;
note ?? 'none'; note!
```

Functions and collections

```
int add(int a, int b) => a + b;
void log({required String id,
    int level = 0}) {}
[1, 2]; {'a': 1}; {1, 2}
[for (final t in trails) t.name,
  if (admin) 'all']
```

Classes, async, records

```
class Trail {
  const Trail(this.id); final String id; }
Future<int> f() async { return await g(); }
(String, int) pair = ('a', 1);
'${trail.name}: $count'
```

Sealed and switch

```
switch (r) {
  Success(:final value) => value,
  Failure() => null }
```

WIDGETS — LAYOUT

Common widgets

```
MaterialApp  Scaffold  AppBar  Text  Icon
ElevatedButton  TextButton  IconButton
FloatingActionButton
TextField  Form  TextFormField   # input
ListView.builder  GridView.builder  # lazy
Card  ListTile  Chip  Dialog  SnackBar
```

Layout

```
Row, Column     # main/crossAxisAlignment
Expanded, Flexible  # share leftover space
Stack + Positioned  # overlay children
Padding  SizedBox  Center  Align
Container  DecoratedBox  # size, colour
SingleChildScrollView  SafeArea
```

Rule

Constraints go down, sizes go up, parents set positions. Bases: StatelessWidget, StatefulWidget, ConsumerWidget.

STATE — RIVERPOD

Notifier

```
class FavouritesNotifier
    extends Notifier<Set<String>> {
  @override
  Set<String> build() => {};
  void add(String id) =>
      state = {...state, id};
}
```

Declare and use

```
final favouritesProvider =
  NotifierProvider<FavouritesNotifier,
    Set<String>>(FavouritesNotifier.new);
ref.watch(p); ref.read(p); ref.listen(...)
```

Notes

Wrap the app in ProviderScope. AsyncNotifierProvider exposes AsyncValue (loading, data, error). Replace state with a new value, never mutate it. Local state: setState, ValueNotifier.

ROUTING — GO_ROUTER

Routes and guard

```
GoRouter(
  initialLocation: '/trails',
  redirect: (context, state) =>
      signedIn ? null : '/login',
  routes: [GoRoute(
    path: '/trails',
    builder: (context, state) => ...,
    routes: [GoRoute(path: ':id', ...)])])
```

Navigate

```
context.go('/trails/42'); // replace stack
context.push('/log');     // push on stack
context.pop();            // go back
```

Parameters and tabs

```
state.pathParameters['id']
state.uri.queryParameters['q']
StatefulShellRoute.indexedStack
// bottom tabs, each with its own stack
```

NETWORK — DIO & STORAGE

Configure dio once

```
final dio = Dio(BaseOptions(
  baseUrl: 'https://api.example.com/v1',
  connectTimeout:
      const Duration(seconds: 10)));
dio.interceptors
    .add(AuthInterceptor(dio, tokens));
```

GET and decode

```
final r = await dio
    .get<List<dynamic>>('/trails');
Trail.fromJson(
    json as Map<String, dynamic>)
// small apps: http.get(Uri), http.post
```

Local storage

```
final prefs =
    await SharedPreferences.getInstance();
await getApplicationDocumentsDirectory();
// drift: tables and watch() streams
```

PLATFORM — INTEGRATE

MethodChannel

```
const _channel = MethodChannel(
    'com.example.trailhead/battery');
await _channel
    .invokeMethod<int>('getBatteryLevel');
// throws PlatformException on failure
```

Location and camera

```
await Geolocator.checkPermission();
await Geolocator.requestPermission();
final cameras = await availableCameras();
CameraController(cameras.first,
    ResolutionPreset.medium)
```

Permissions

Declare in AndroidManifest.xml and Info.plist (NSCameraUsageDescription, ...); a missing iOS key crashes on use. Pigeon generates both ends of a channel.

TESTING — VERIFY

Commands

```
flutter test             # unit + widget
flutter test --coverage
# writes coverage/lcov.info
flutter test --update-goldens
flutter test integration_test # device
```

Widget test

```
testWidgets('...', (tester) async {
  await tester.pumpWidget(const App());
  await tester
      .tap(find.byType(IconButton));
  await tester.pumpAndSettle();
  expect(find.text('Ridge'),
      findsOneWidget);
});
```

Fakes

Wrap the widget in ProviderScope with overrides to inject fakes; mocktail 1.0.5 for mocks.

BUILD — RELEASE

Artifacts

```
flutter build apk --release
flutter build appbundle --release
# Android App Bundle for Google Play
flutter build ipa --release
# iOS archive for App Store Connect
```

Options

```
--dart-define=KEY=value
--dart-define-from-file=file.json
--obfuscate
  --split-debug-info=build/symbols
--analyze-size
```

Version in pubspec.yaml

```
version: 1.4.0+27  # name + build number
```